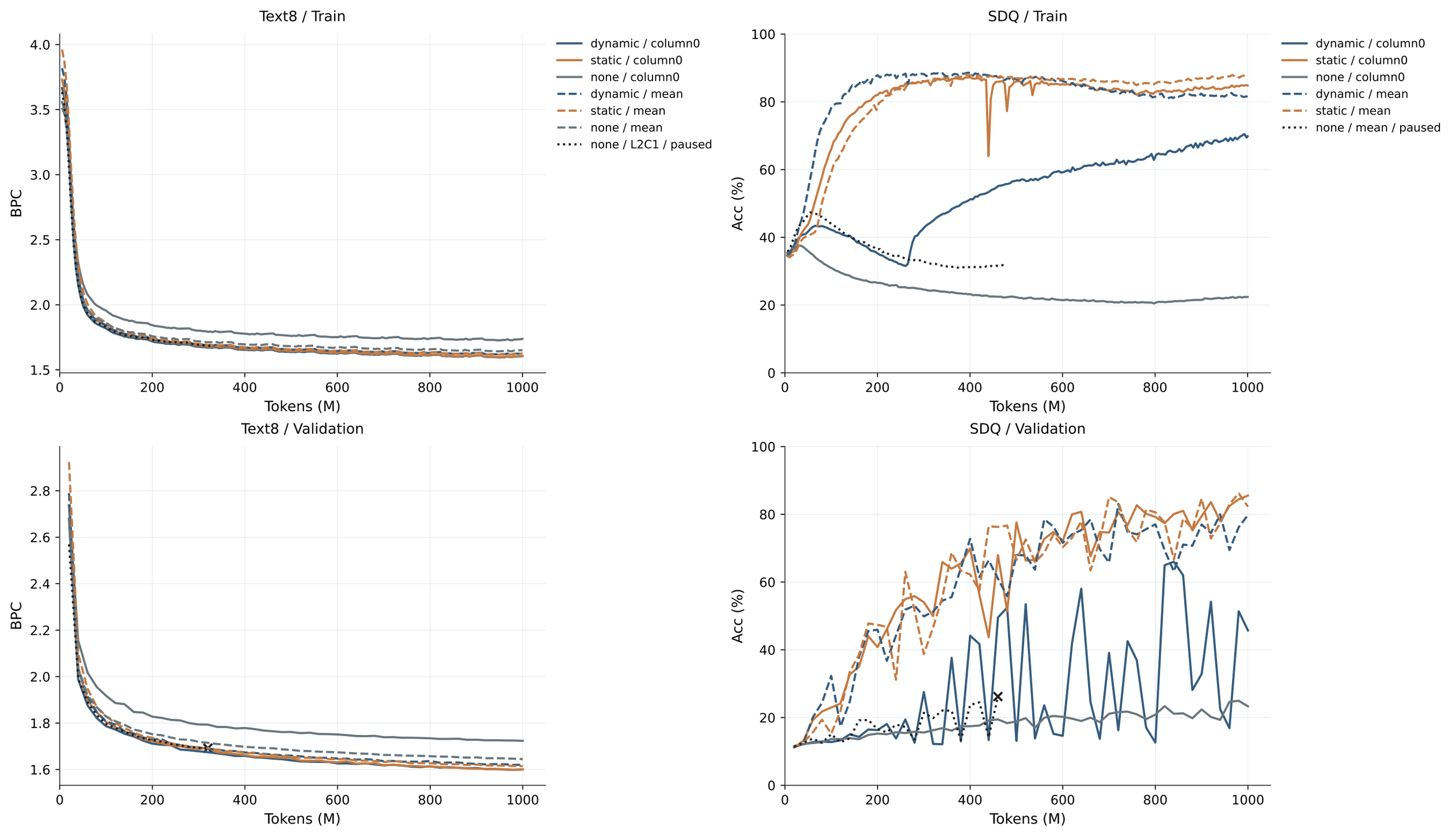
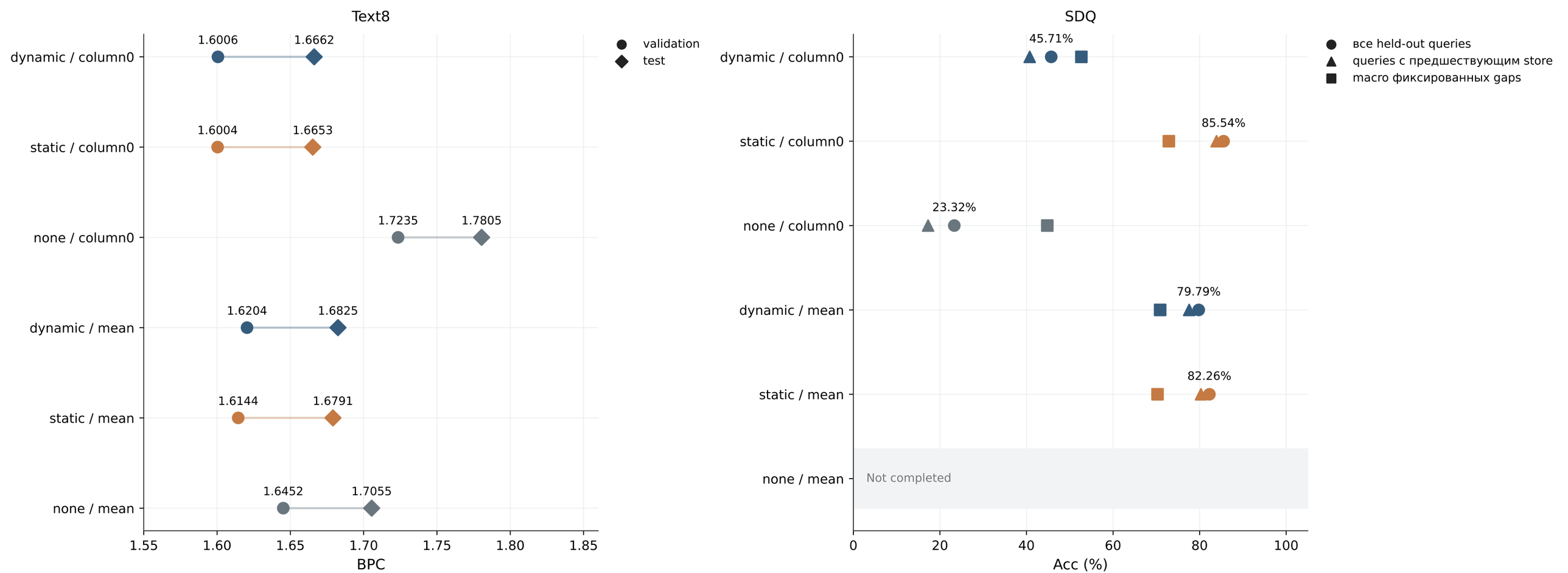
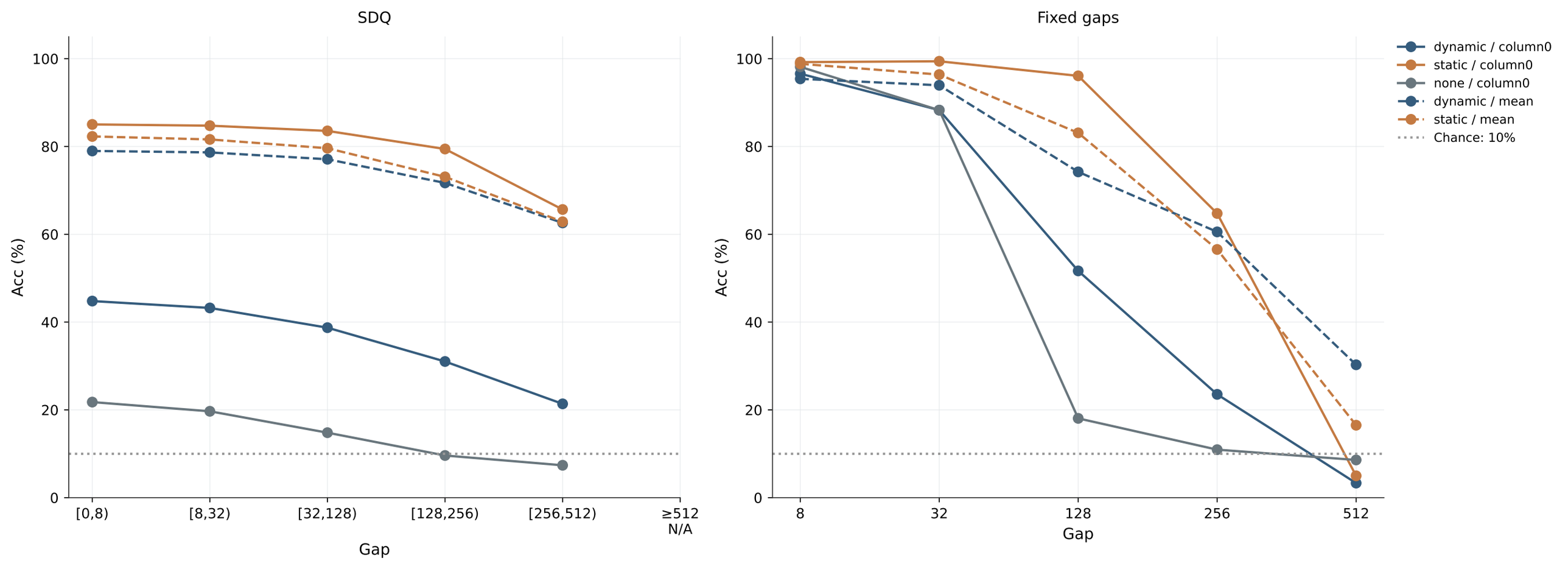
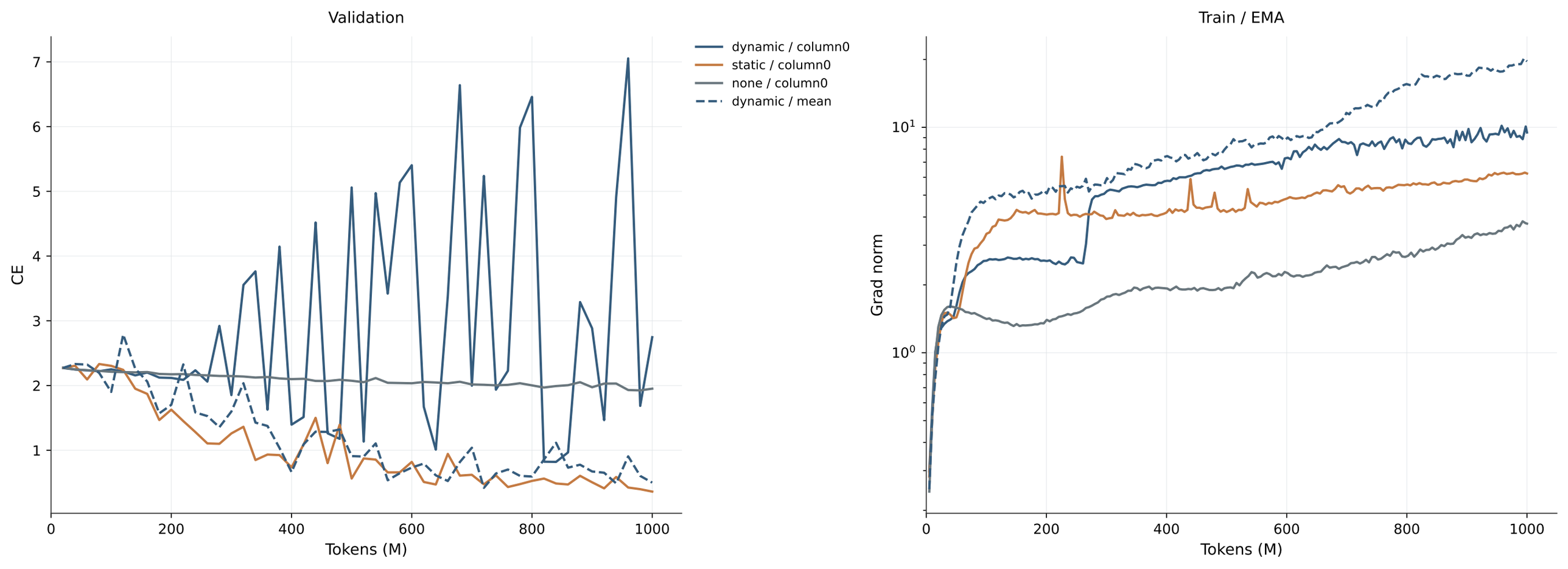
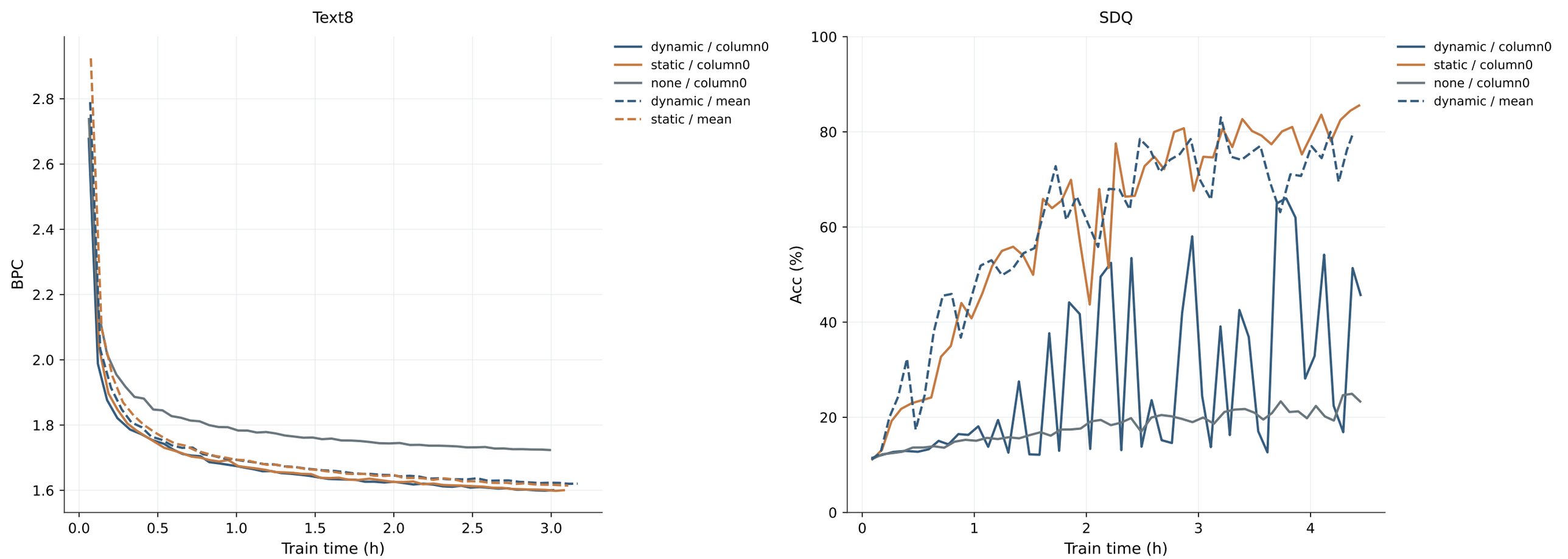
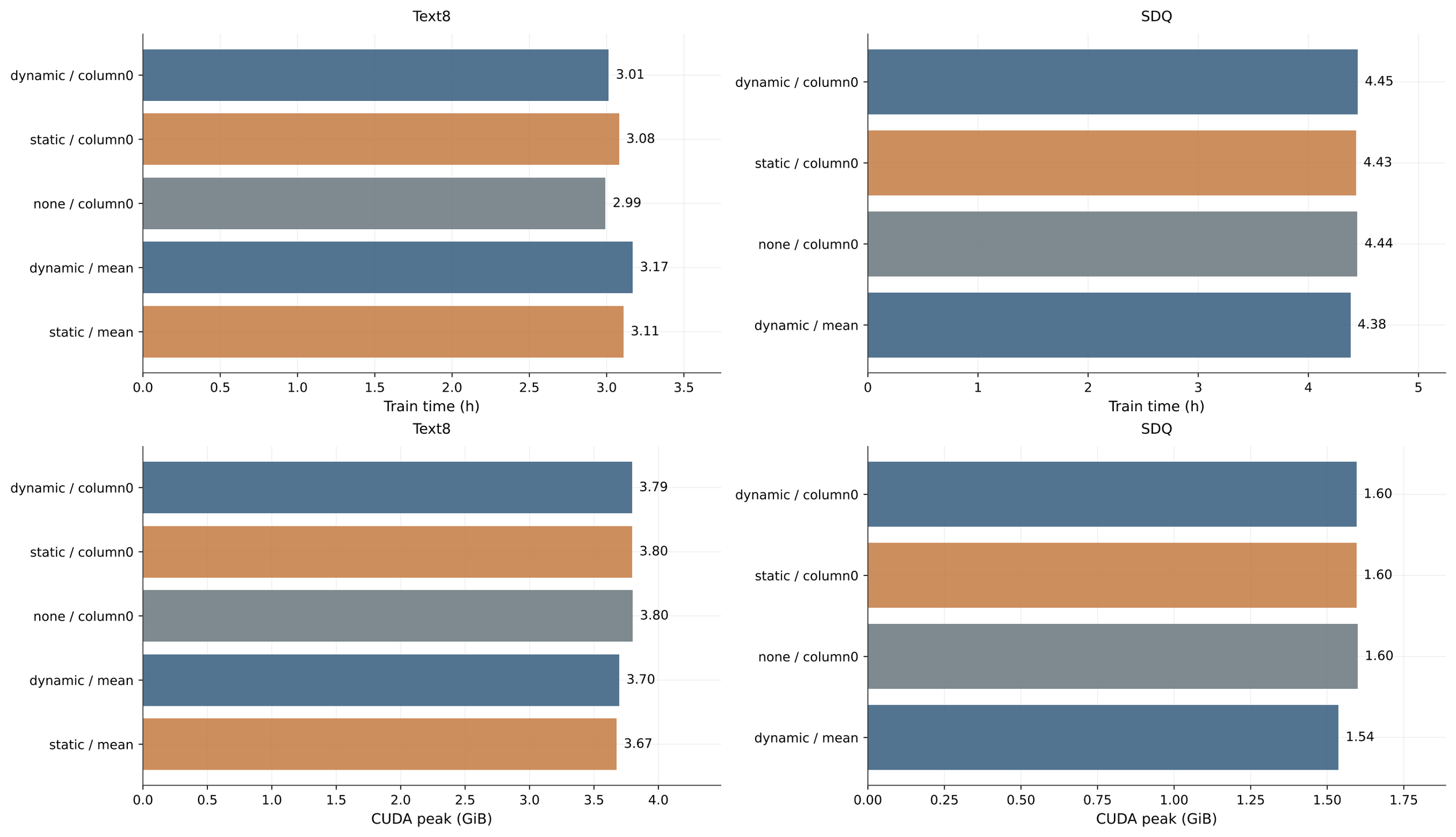

Text8 / Train
BPC
Tokens (M)
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
none / mean
none / L2C1 / paused
SDQ / Train
Acc (%)
Tokens (M)
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
none / mean / paused
Text8 / Validation
BPC
Tokens (M)
SDQ / Validation
Acc (%)
Tokens (M)

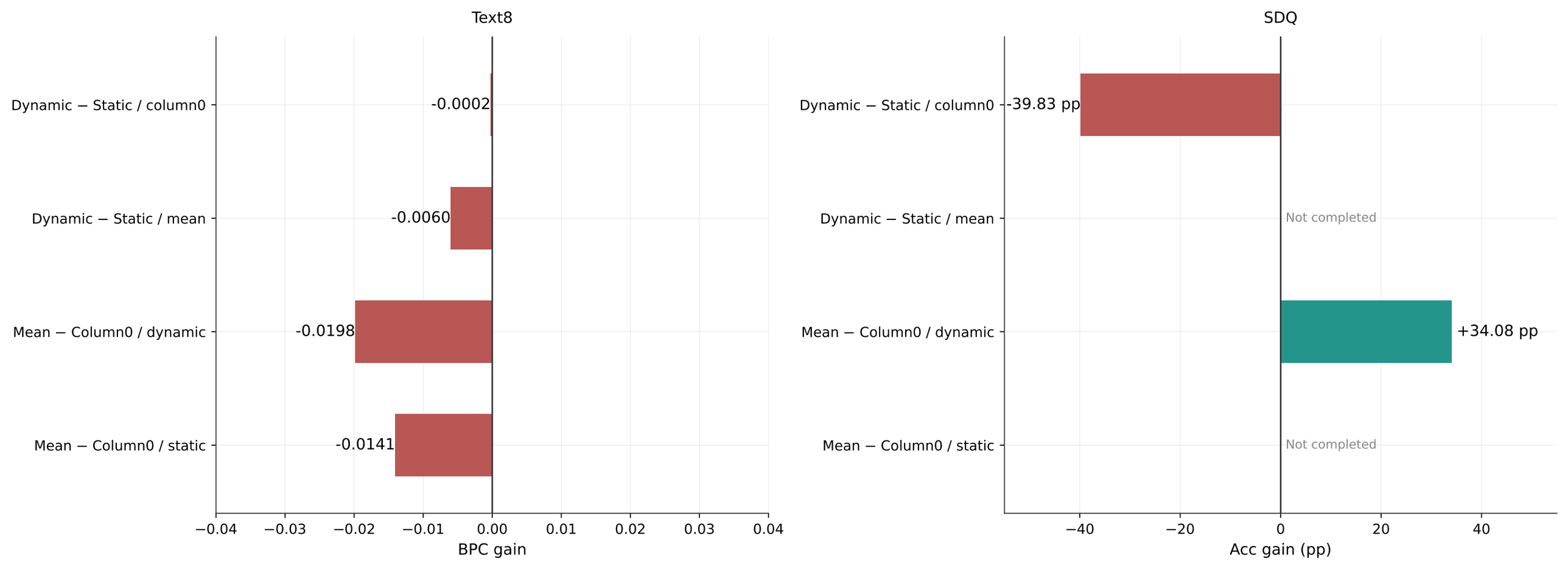

Text8
validation
test
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
none / mean
1.6006
1.6662
1.6004
1.6653
1.7235
1.7805
1.6204
1.6825
1.6144
1.6791
1.6452
1.7055
1.55
1.60
1.65
1.70
1.75
1.80
1.85
BPC
SDQ
все held-out queries
queries с предшествующим store
macro фиксированных gaps
45.71%
85.54%
23.32%
79.79%
82.26%
Not completed
0
20
40
60
80
100
Acc (%)

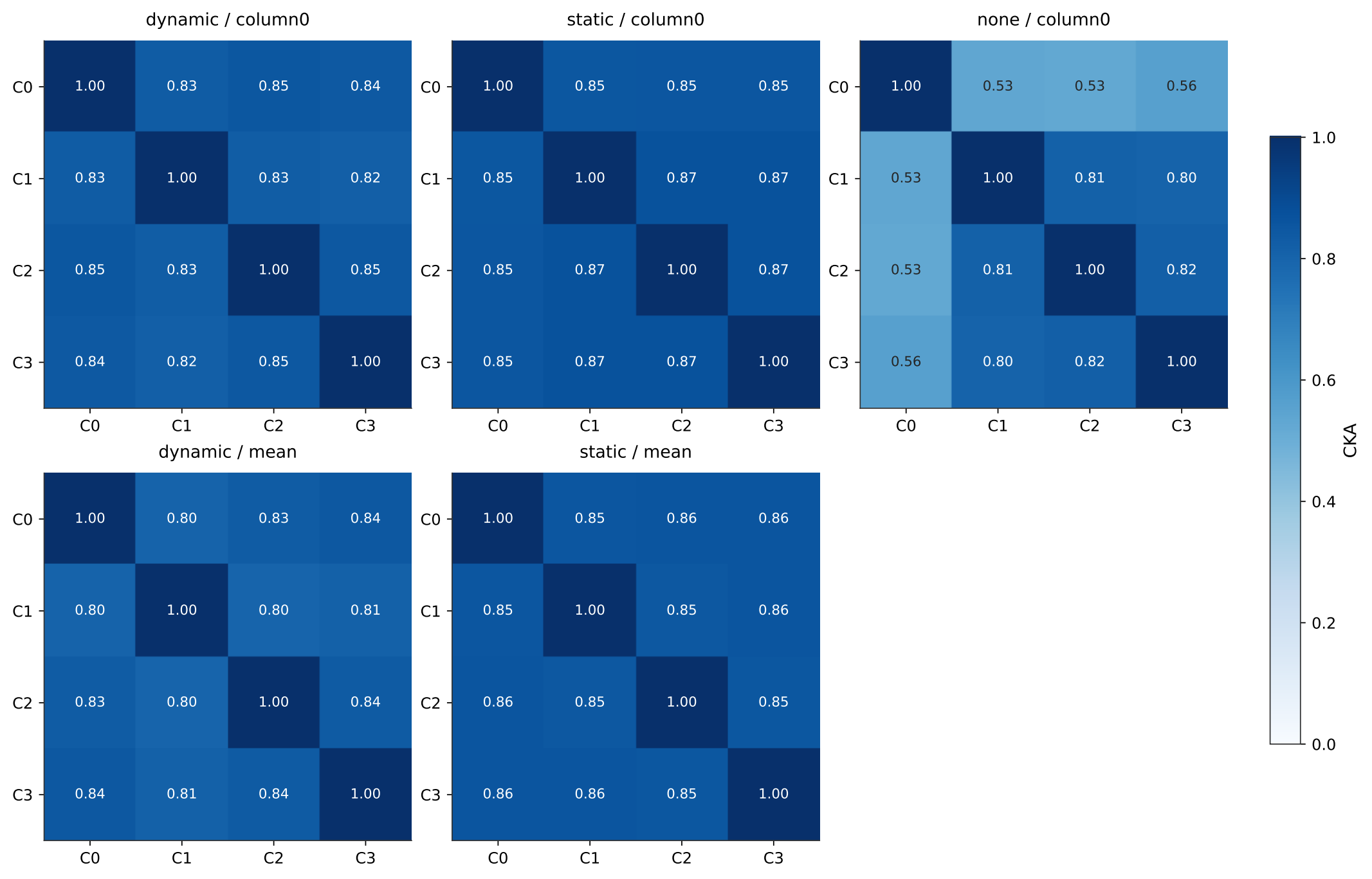

SDQ
Fixed gaps
Acc (%)
Gap
[0,8)
[8,32)
[32,128)
[128,256)
[256,512)
≥512
N/A
8
32
128
256
512
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
Chance: 10%

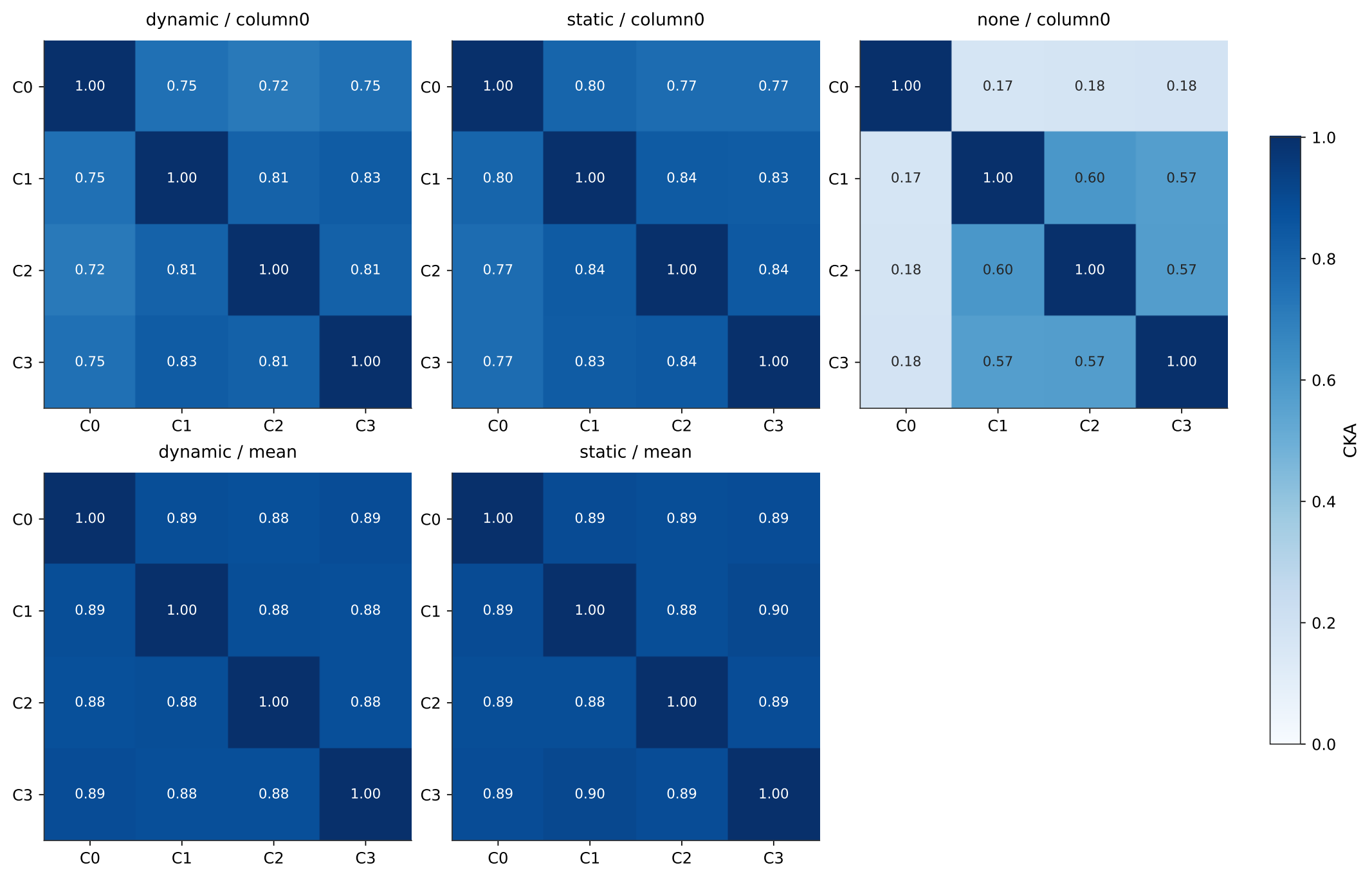

Validation
CE
Tokens (M)
dynamic / column0
static / column0
none / column0
dynamic / mean
Train / EMA
Grad norm
Tokens (M)

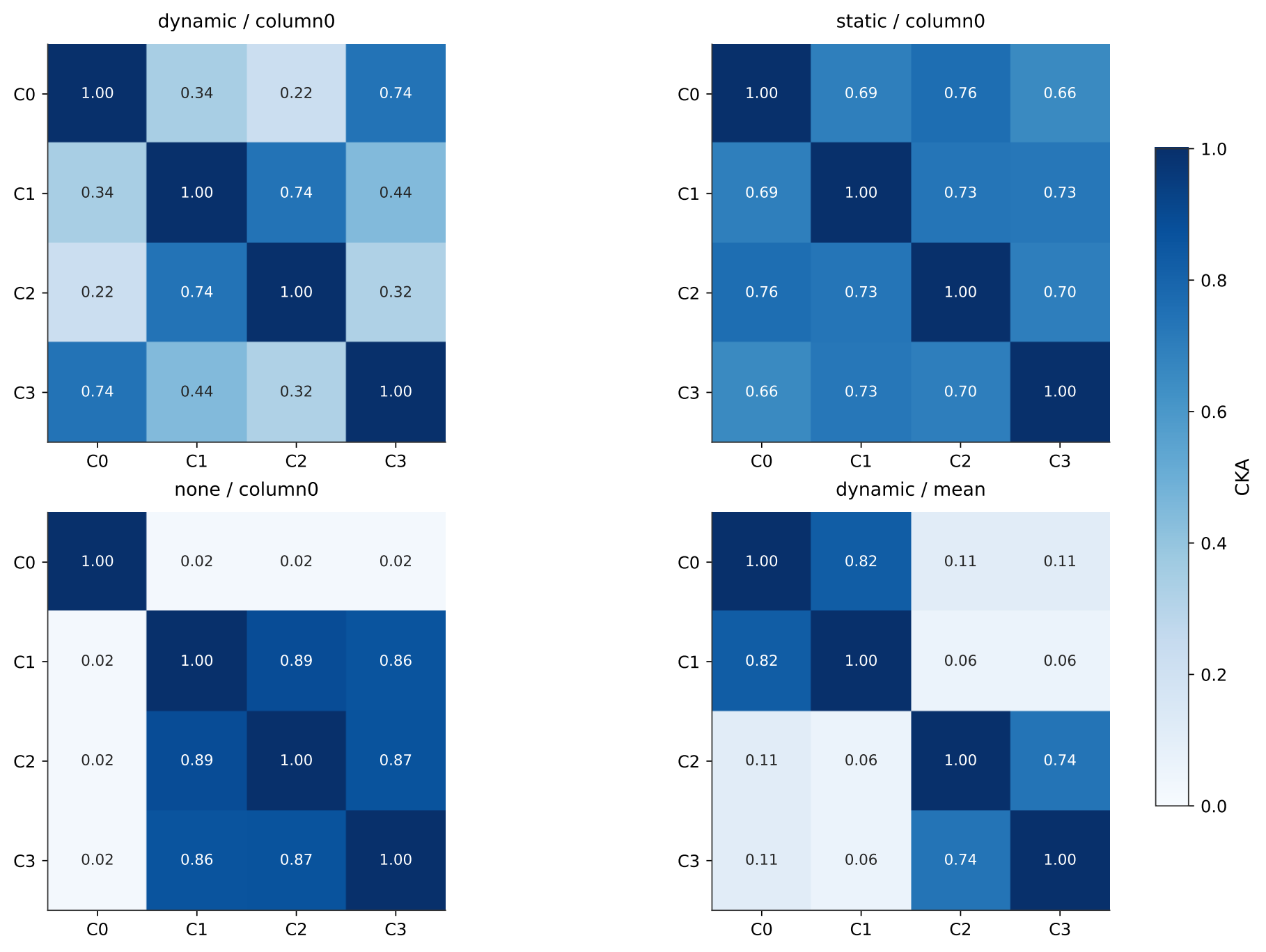

Text8
BPC
Train time (h)
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
SDQ
Acc (%)
Train time (h)
dynamic / column0
static / column0
none / column0
dynamic / mean

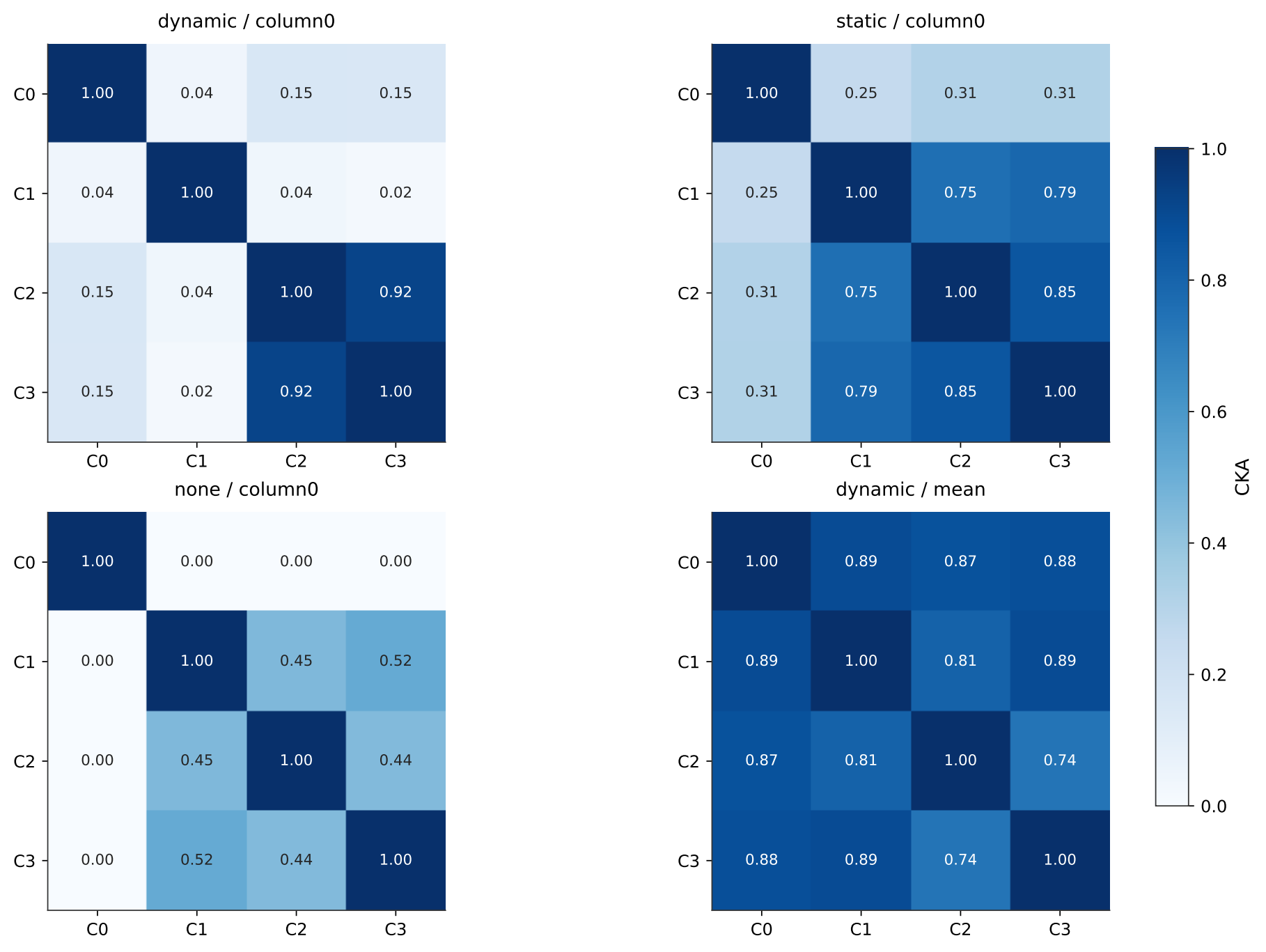

Text8
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
3.01
3.08
2.99
3.17
3.11
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Train time (h)
SDQ
dynamic / column0
static / column0
none / column0
dynamic / mean
4.45
4.43
4.44
4.38
0
1
2
3
4
5
Train time (h)
Text8
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
3.79
3.80
3.80
3.70
3.67
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
CUDA peak (GiB)
SDQ
dynamic / column0
static / column0
none / column0
dynamic / mean
1.60
1.60
1.60
1.54
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
CUDA peak (GiB)

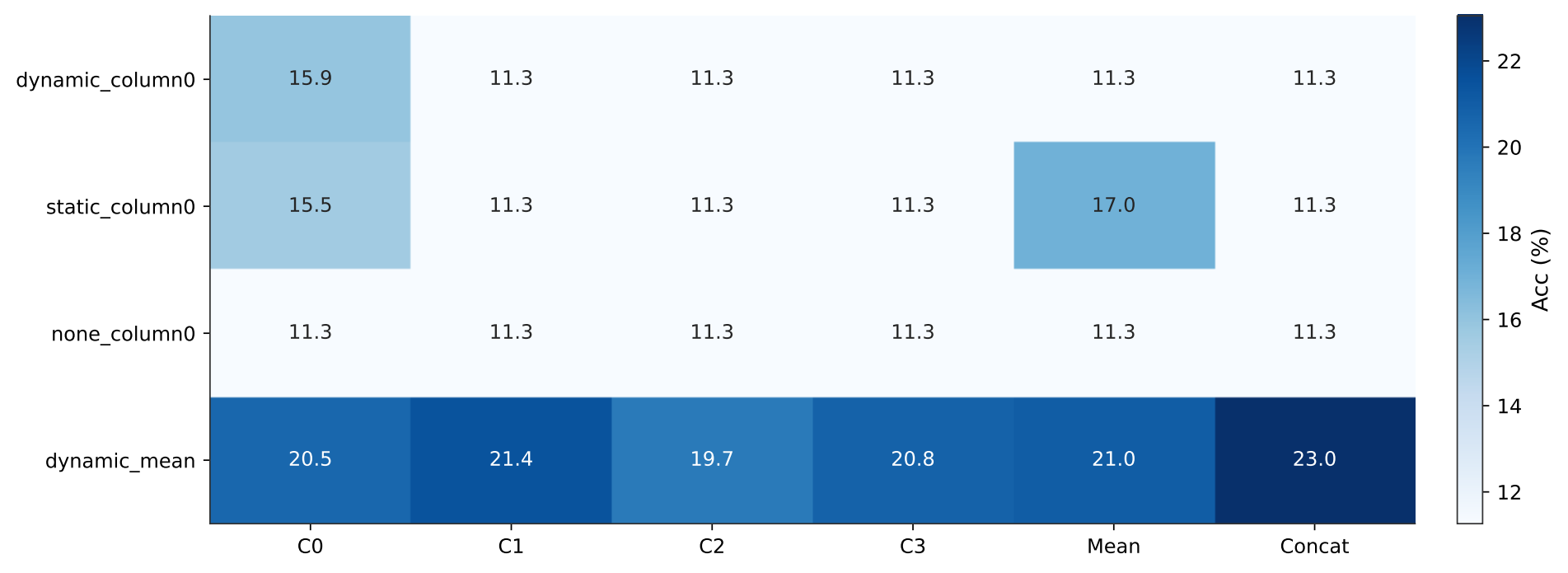

Text8
Dynamic − Static / column0
Dynamic − Static / mean
Mean − Column0 / dynamic
Mean − Column0 / static
-0.0002
-0.0060
-0.0198
-0.0141
−0.04
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
BPC gain
SDQ
Dynamic − Static / column0
Dynamic − Static / mean
Mean − Column0 / dynamic
Mean − Column0 / static
-39.83 pp
Not completed
+34.08 pp
Not completed
−40
−20
0
20
40
Acc gain (pp)

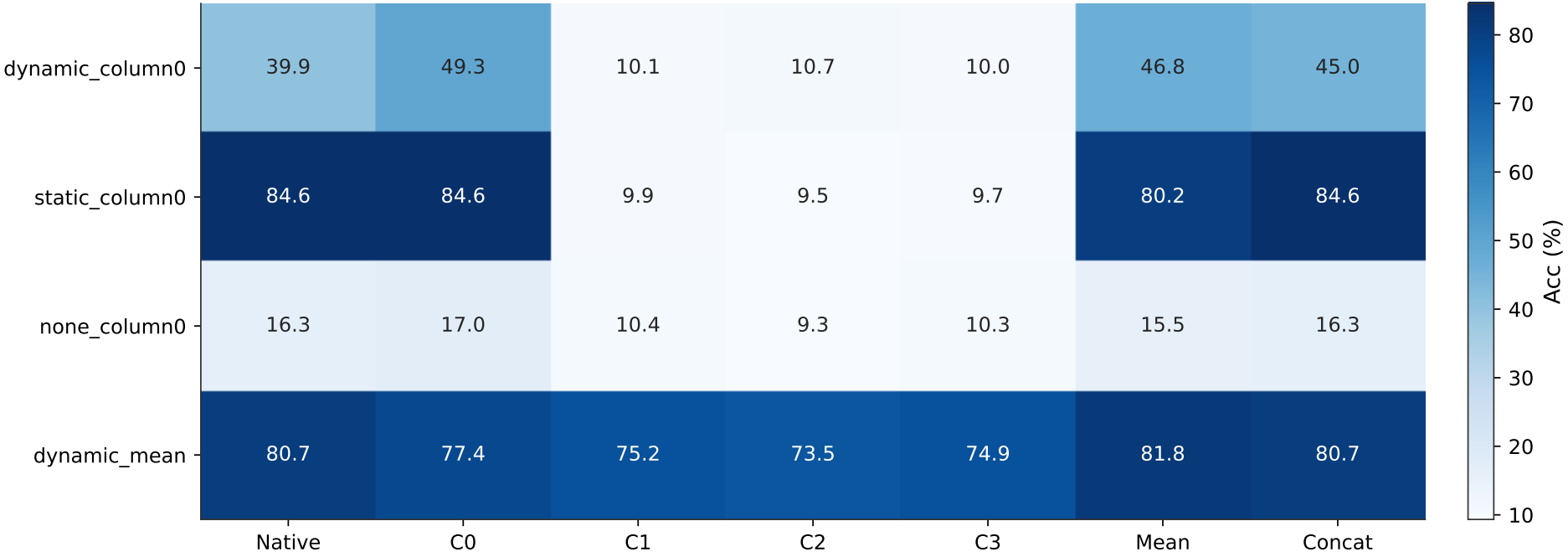

dynamic / column0
C0 1.00 0.83 0.85 0.84
C1 0.83 1.00 0.83 0.82
C2 0.85 0.83 1.00 0.85
C3 0.84 0.82 0.85 1.00
C0 C1 C2 C3
static / column0
C0 1.00 0.85 0.85 0.85
C1 0.85 1.00 0.87 0.87
C2 0.85 0.87 1.00 0.87
C3 0.85 0.87 0.87 1.00
C0 C1 C2 C3
none / column0
C0 1.00 0.53 0.53 0.56
C1 0.53 1.00 0.81 0.80
C2 0.53 0.81 1.00 0.82
C3 0.56 0.80 0.82 1.00
C0 C1 C2 C3
dynamic / mean
C0 1.00 0.80 0.83 0.84
C1 0.80 1.00 0.80 0.81
C2 0.83 0.80 1.00 0.84
C3 0.84 0.81 0.84 1.00
C0 C1 C2 C3
static / mean
C0 1.00 0.85 0.86 0.86
C1 0.85 1.00 0.85 0.86
C2 0.86 0.85 1.00 0.85
C3 0.86 0.86 0.85 1.00
C0 C1 C2 C3
CKA
1.0
0.8
0.6
0.4
0.2
0.0

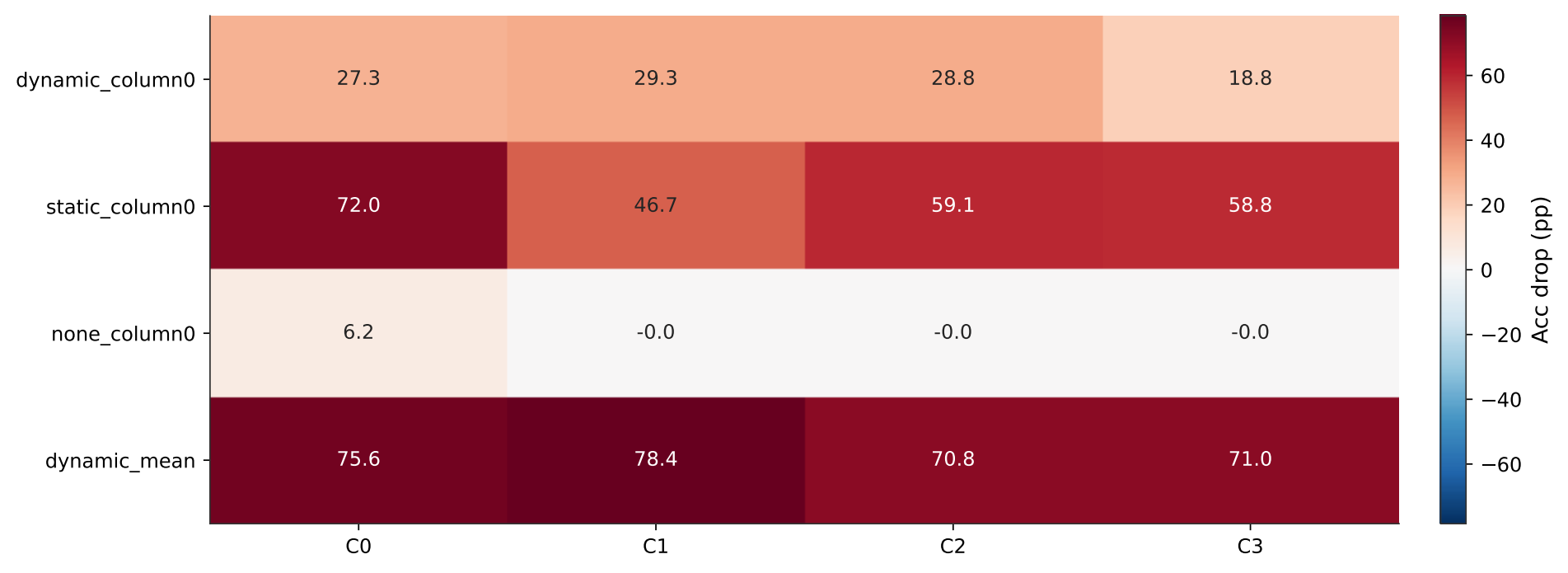

dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
C0
C1
C2
C3
1.00 0.75 0.72 0.75
0.75 1.00 0.81 0.83
0.72 0.81 1.00 0.81
0.75 0.83 0.81 1.00
1.00 0.80 0.77 0.77
0.80 1.00 0.84 0.83
0.77 0.84 1.00 0.84
0.77 0.83 0.84 1.00
1.00 0.17 0.18 0.18
0.17 1.00 0.60 0.57
0.18 0.60 1.00 0.57
0.18 0.57 0.57 1.00
1.00 0.89 0.88 0.89
0.89 1.00 0.88 0.88
0.88 0.88 1.00 0.88
0.89 0.88 0.88 1.00
1.00 0.89 0.89 0.89
0.89 1.00 0.88 0.90
0.89 0.88 1.00 0.89
0.89 0.90 0.89 1.00
CKA
1.0
0.8
0.6
0.4
0.2
0.0

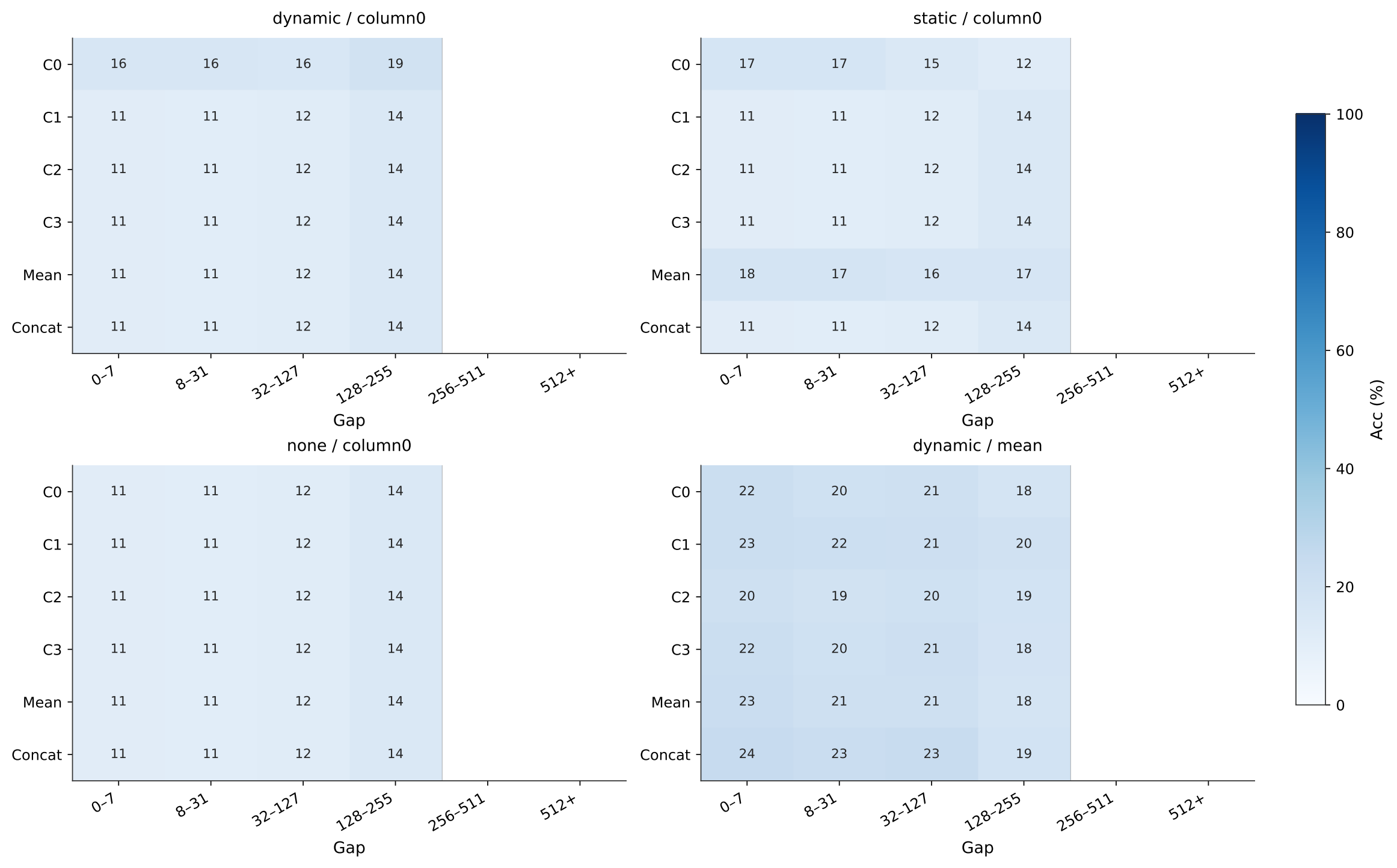

dynamic / column0
C0 1.00 0.34 0.22 0.74
C1 0.34 1.00 0.74 0.44
C2 0.22 0.74 1.00 0.32
C3 0.74 0.44 0.32 1.00
C0 C1 C2 C3
static / column0
C0 1.00 0.69 0.76 0.66
C1 0.69 1.00 0.73 0.73
C2 0.76 0.73 1.00 0.70
C3 0.66 0.73 0.70 1.00
C0 C1 C2 C3
none / column0
C0 1.00 0.02 0.02 0.02
C1 0.02 1.00 0.89 0.86
C2 0.02 0.89 1.00 0.87
C3 0.02 0.86 0.87 1.00
C0 C1 C2 C3
dynamic / mean
C0 1.00 0.82 0.11 0.11
C1 0.82 1.00 0.06 0.06
C2 0.11 0.06 1.00 0.74
C3 0.11 0.06 0.74 1.00
C0 C1 C2 C3
CKA
1.0
0.8
0.6
0.4
0.2
0.0

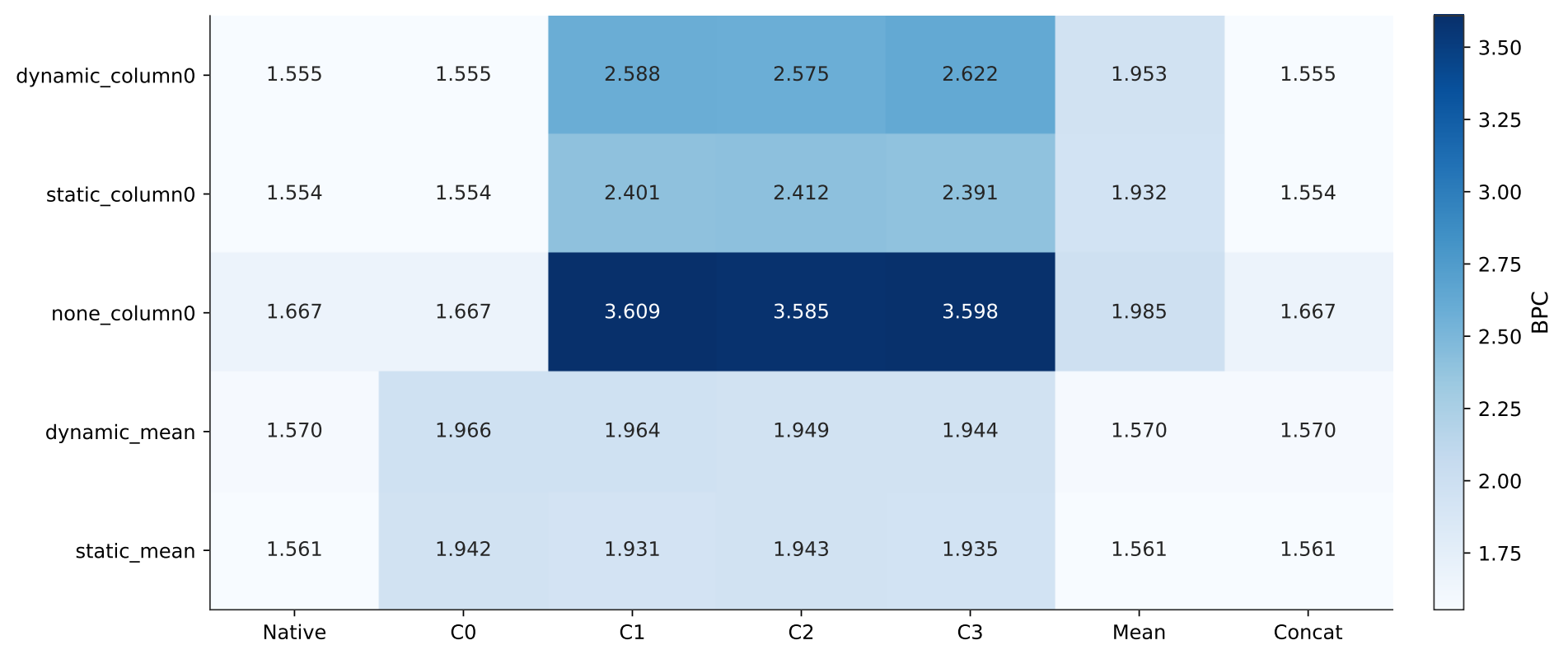

dynamic / column0
C0 1.00 0.04 0.15 0.15
C1 0.04 1.00 0.04 0.02
C2 0.15 0.04 1.00 0.92
C3 0.15 0.02 0.92 1.00
C0 C1 C2 C3
static / column0
C0 1.00 0.25 0.31 0.31
C1 0.25 1.00 0.75 0.79
C2 0.31 0.75 1.00 0.85
C3 0.31 0.79 0.85 1.00
C0 C1 C2 C3
none / column0
C0 1.00 0.00 0.00 0.00
C1 0.00 1.00 0.45 0.52
C2 0.00 0.45 1.00 0.44
C3 0.00 0.52 0.44 1.00
C0 C1 C2 C3
dynamic / mean
C0 1.00 0.89 0.87 0.88
C1 0.89 1.00 0.81 0.89
C2 0.87 0.81 1.00 0.74
C3 0.88 0.89 0.74 1.00
C0 C1 C2 C3
CKA
1.0
0.8
0.6
0.4
0.2
0.0

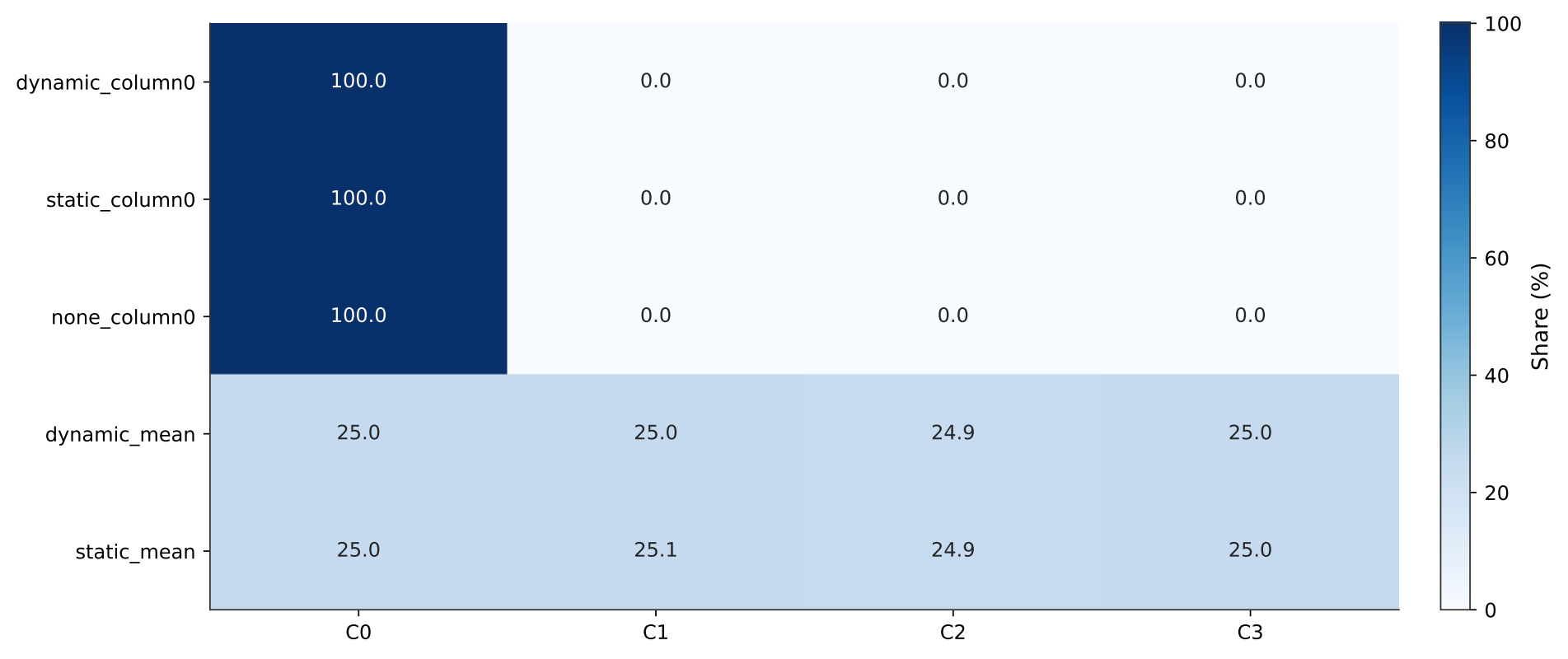

dynamic_column0
static_column0
none_column0
dynamic_mean
15.9
11.3
11.3
11.3
11.3
11.3
15.5
11.3
11.3
11.3
17.0
11.3
11.3
11.3
11.3
11.3
11.3
11.3
20.5
21.4
19.7
20.8
21.0
23.0
C0
C1
C2
C3
Mean
Concat
Acc (%)
22
20
18
16
14
12


dynamic_column0
static_column0
none_column0
dynamic_mean
39.9
49.3
10.1
10.7
10.0
46.8
45.0
84.6
84.6
9.9
9.5
9.7
80.2
84.6
16.3
17.0
10.4
9.3
10.3
15.5
16.3
80.7
77.4
75.2
73.5
74.9
81.8
80.7
Native
C0
C1
C2
C3
Mean
Concat
Acc (%)
80
70
60
50
40
30
20
10


dynamic_column0
static_column0
none_column0
dynamic_mean
27.3
29.3
28.8
18.8
72.0
46.7
59.1
58.8
6.2
-0.0
-0.0
-0.0
75.6
78.4
70.8
71.0
C0
C1
C2
C3
Acc drop (pp)
60
40
20
0
−20
−40
−60

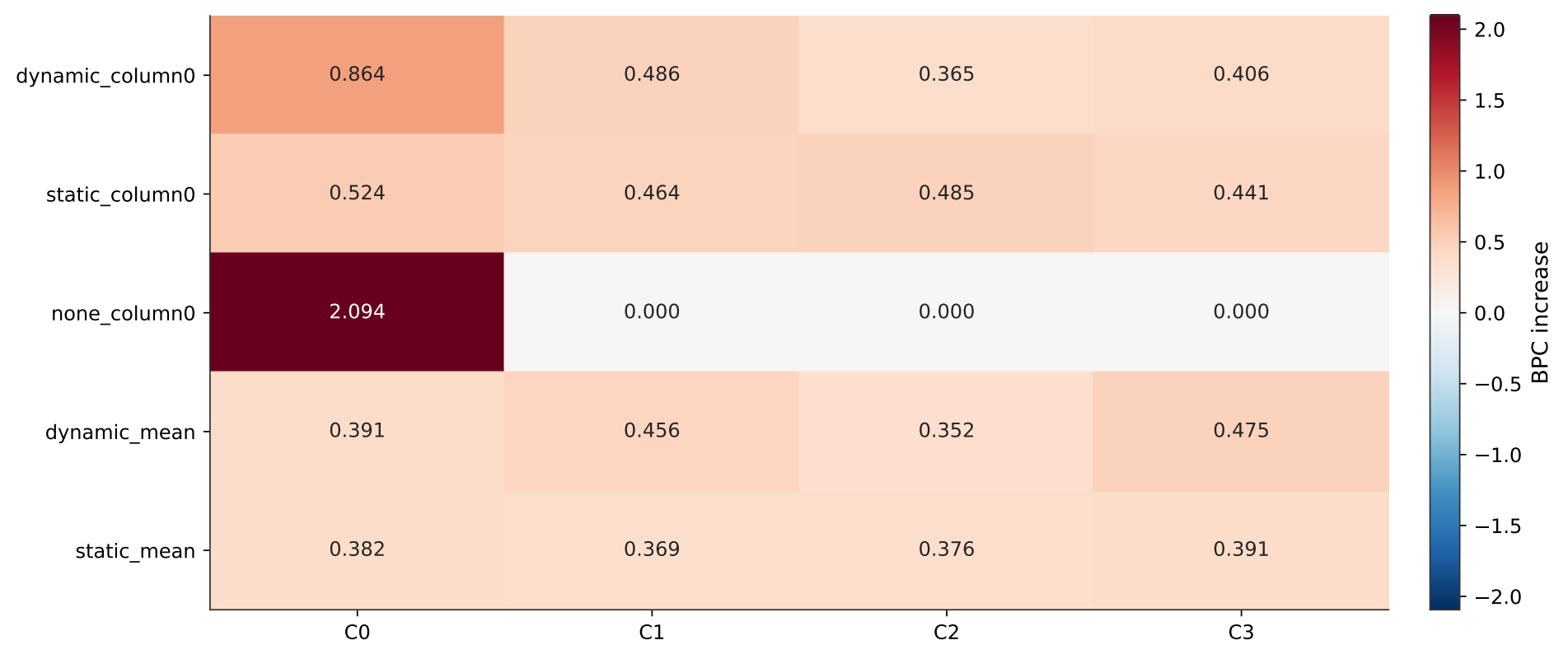

dynamic / column0
C0 16 16 16 19
C1 11 11 12 14
C2 11 11 12 14
C3 11 11 12 14
Mean 11 11 12 14
Concat 11 11 12 14
0–7
8–31
32–127
128–255
256–511
512+
Gap
static / column0
C0 17 17 15 12
C1 11 11 12 14
C2 11 11 12 14
C3 11 11 12 14
Mean 18 17 16 17
Concat 11 11 12 14
0–7
8–31
32–127
128–255
256–511
512+
Gap
none / column0
C0 11 11 12 14
C1 11 11 12 14
C2 11 11 12 14
C3 11 11 12 14
Mean 11 11 12 14
Concat 11 11 12 14
0–7
8–31
32–127
128–255
256–511
512+
Gap
dynamic / mean
C0 22 20 21 18
C1 23 22 21 20
C2 20 19 20 19
C3 22 20 21 18
Mean 23 21 21 18
Concat 24 23 23 19
0–7
8–31
32–127
128–255
256–511
512+
Gap
Acc (%)
100
80
60
40
20
0


dynamic_column0
static_column0
none_column0
dynamic_mean
static_mean
1.555 1.555 2.588 2.575 2.622 1.953 1.555
1.554 1.554 2.401 2.412 2.391 1.932 1.554
1.667 1.667 3.609 3.585 3.598 1.985 1.667
1.570 1.966 1.964 1.949 1.944 1.570 1.570
1.561 1.942 1.931 1.943 1.935 1.561 1.561
Native
C0
C1
C2
C3
Mean
Concat
BPC
3.50
3.25
3.00
2.75
2.50
2.25
2.00
1.75


dynamic_column0
static_column0
none_column0
dynamic_mean
static_mean
100.0
0.0
0.0
0.0
100.0
0.0
0.0
0.0
100.0
0.0
0.0
0.0
25.0
25.0
24.9
25.0
25.0
25.1
24.9
25.0
C0
C1
C2
C3
100
80
60
40
20
0
Share (%)


dynamic_column0
static_column0
none_column0
dynamic_mean
static_mean
2.848
0.000
0.000
0.000
2.856
0.000
0.000
0.000
2.726
0.000
0.000
0.000
0.131
0.148
0.139
0.138
0.137
0.175
0.155
0.172
C0
C1
C2
C3
BPC increase
2
1
0
−1
−2


dynamic_column0
static_column0
none_column0
dynamic_mean
34.7
-1.3
-0.9
-1.3
73.6
-0.0
-0.0
-0.0
6.1
-0.0
-0.0
-0.0
4.3
2.4
3.6
2.3
C0
C1
C2
C3
60
40
20
0
−20
−40
−60
Acc drop (pp)


dynamic_column0
static_column0
none_column0
dynamic_mean
static_mean
0.864
0.486
0.365
0.406
0.524
0.464
0.485
0.441
2.094
0.000
0.000
0.000
0.391
0.456
0.352
0.475
0.382
0.369
0.376
0.391
C0
C1
C2
C3
BPC increase
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0